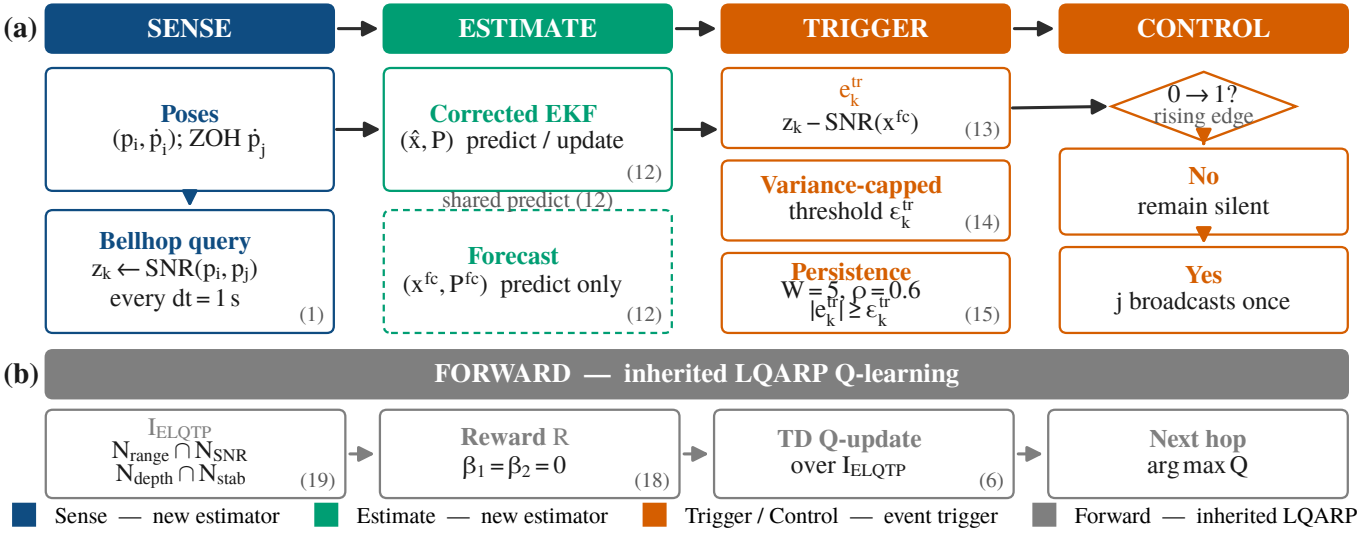
(a)
SENSE
ESTIMATE
TRIGGER
CONTROL
Poses
(p_i, ṗ_i); ZOH ṗ_j
Bellhop query
z_k ← SNR(p_i, p_j)
every dt = 1 s
(1)
Corrected EKF
(x̂, P) predict / update
(12)
shared predict (12)
Forecast
(x^fc, P^fc) predict only
(12)
e_k^tr
z_k − SNR(x^fc)
(13)
Variance-capped
threshold ε_k^tr
(14)
Persistence
W = 5, ρ = 0.6
|e_k^tr| ≥ ε_k^tr
(15)
0 → 1?
rising edge
No
remain silent
Yes
j broadcasts once
(b)
FORWARD — inherited LQARP Q-learning
I_ELQTP
N_range ∩ N_SNR
N_depth ∩ N_stab
(19)
Reward R
β_1 = β_2 = 0
(18)
TD Q-update
over I_ELQTP
(6)
Next hop
arg max Q
Sense — new estimator
Estimate — new estimator
Trigger / Control — event trigger
Forward — inherited LQARP